

Class of 2029

PPOM I OMM Lab #29

**High-Velocity Low-Amplitude (HVLA)
for the Lower Extremities
(Tibia, Fibula, Subtalar, Tarsals and
Metatarsal regions)**

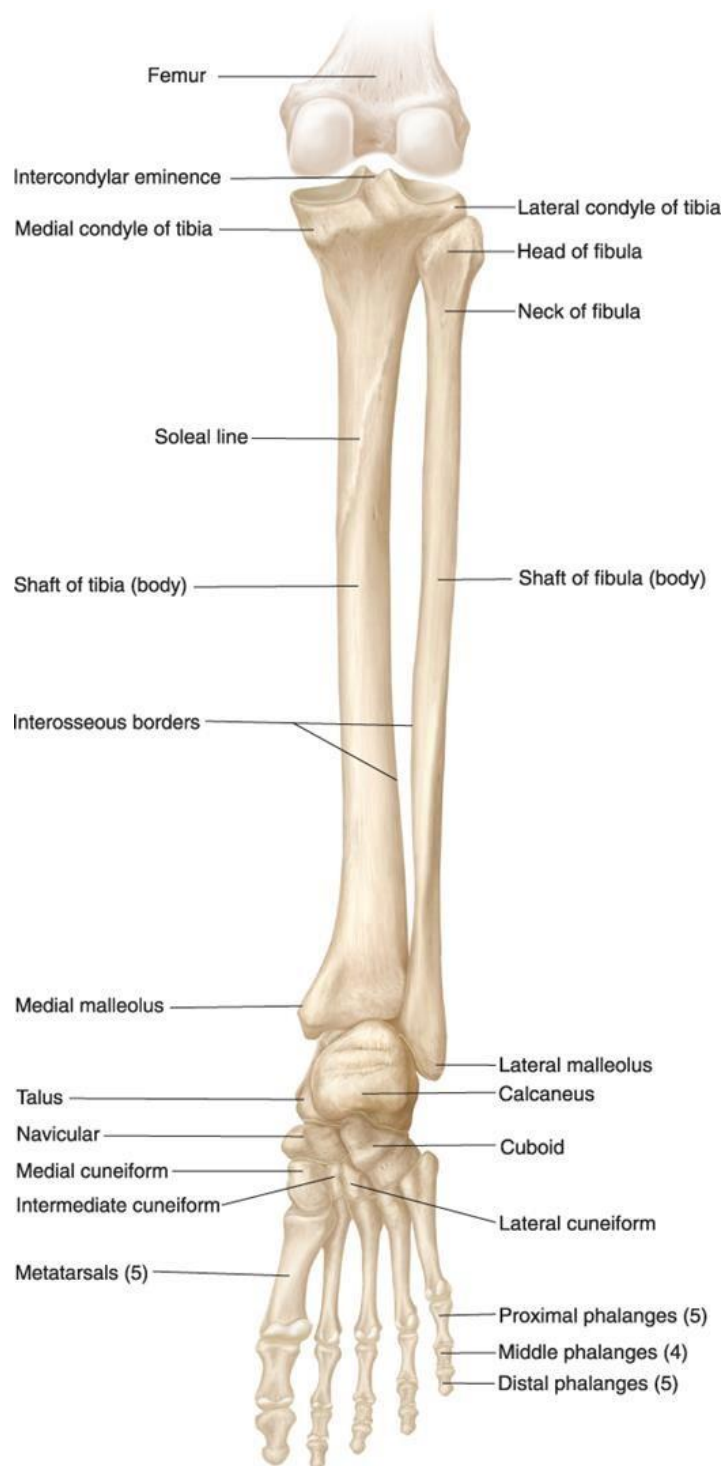
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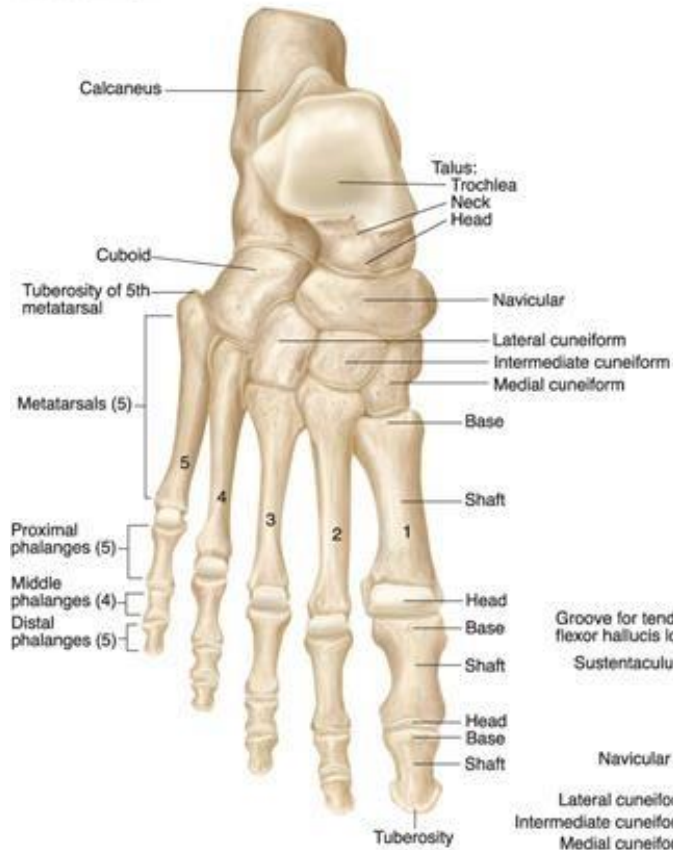
Anatomy of the Distal Lower Limb



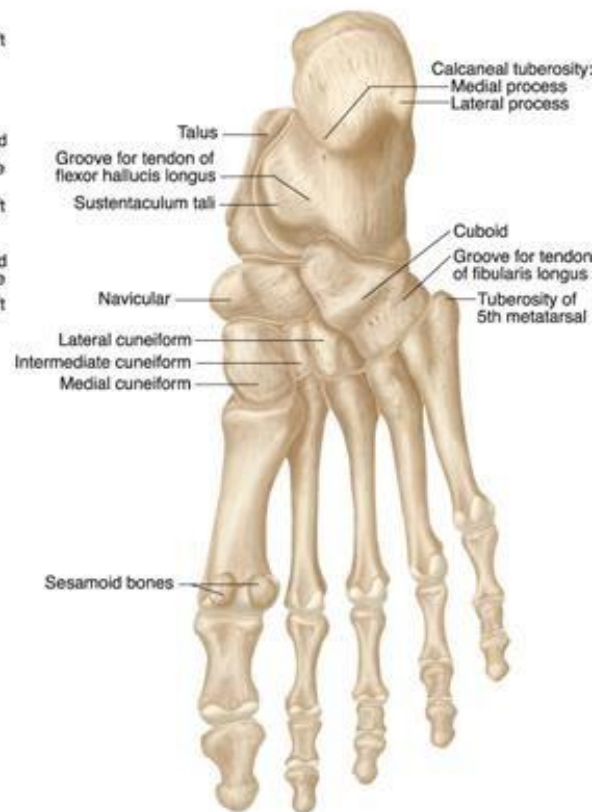
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Anatomy of the Foot

A. Dorsal view



B. Plantar view



<https://login.nyit.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/content.aspx?sectionid=249368759&bookid=2991>

Lower Extremity Diagnosis

	Patient 1		Patient 2	
	Left	Right	Left	Right
Tibiotalar joint (i.e. tibia on talus)				
Subtalar joint (anterior medial talus vs. posterolateral talus)				
Proximal tibia (i.e. tibia on femur)				
Other (e.g. dropped navicular/cuboid/cuneiforms)				

High-Velocity Low-Amplitude (HVLA) for Dropped Navicular/Cuboid/Cuneiforms (Hiss Whip)

1. The patient is asked to lie prone on the table, with the physician standing at the side of the table closer to the foot of the table and on the side of the somatic dysfunction.
2. The physician flexes the patient's hip and knee on the side of the dysfunction and drops the leg on the side of the dysfunction off the side of the table:



Fig. 11.177

3. The physician uses both hands to grasp the patient's foot, placing their thumbs in a "V" shape over the plantar surface of the cuboid, navicular or cuneiforms, whichever it is that is in a dropped position relative to the other metatarsals.
4. The physician delivers a downward thrust* through their thumbs into the sole of the foot at the level of the dysfunctional cuboid/navicular/cuneiform while simultaneously inducing a whip-like action at the patient's ankle and knee (hence, the name "Hiss Whip" technique).



Fig. 11.178

** The physician can introduce a more medial thrust for the medial tarsals and a more lateral thrust for the lateral tarsals (the lateral aspect of the foot is opened by adducting the forefoot).*

5. The physician reassesses the range of motion of the forefoot and palpates for the dropped cuboid/navicular/cuneiform to determine effectiveness of treatment.

[Technique #10 in LE Region Section \(Video 11.36\) in Nicholas & Nicholas](#)

HVLA Anterior Tibiotalar Dysfunction (Anterior Tibia on Talus; Posterior Talus)

1. The patient is asked to lie supine on the table, with the physician standing at the foot of the table.
2. The physician uses their caudad hand to cup the calcaneus anchoring the foot (here, slight traction may be applied).
3. The physician places their cephalad hand on the tibia proximal to the ankle mortise:



Fig. 11.172

4. The physician delivers a thrust through their cephalad hand (on the tibia) directed straight down toward the table (white arrow in the image below):



Fig. 11.173

5. The physician reassesses the glide of the tibia on the talus and the range of motion of the ankle to determine effectiveness of treatment.

[Technique #8 in LE Region section \(video 11.33\) in Nicholas & Nicholas](#)

HVLA Posterior Tibiotalar Disfunction (Posterior Tibia on Talus; Anterior Talus)

1. The patient is asked to lie supine on the table, with the physician standing at the foot of the table.
2. The physician wraps their hands around the foot with their fingers interlaced on the dorsum of the foot.
3. The physician uses pressure from their thumbs at the ball of the foot to dorsiflex the foot to the motion barrier:



Fig. 11.174

4. The physician performs a gentle dorsiflexion of the foot:



Fig. 11.175

5. While increasing the degree of the foot dorsiflexion, the physician delivers a tractional thrust (white arrow in image below):



Fig. 11.176

6. The physician reassesses the glide of the tibia on the talus and the range of motion of the ankle to determine effectiveness of treatment.

[Technique #9 in LE Region section \(Videos 11.33, 11.34, and 11.35 \) in Nicholas & Nicholas](#)

HVLA Anterior Tibia on the Femur somatic dysfunction

1. The patient is asked to sit on the side of the table with a small pillow beneath the thigh on the side of the dysfunction as a cushion; the physician is seated at the side of the table facing the patient.
2. The physician places their thumbs on the anterior tibial plateau with the fingers wrapping around the leg:



Fig. 11.156

3. The physician ensures the patient's thigh and leg are relaxed by springing the thigh up and down:



Fig. 11.157

4. The physician delivers a thrust straight down toward the floor (traction down), while simultaneously applying a posterior pressure impulse with the thumbs (multidirectional thrust; white arrows in the image below):



Fig. 11.158

5. The physician reassesses anterior glide and the range of motion of the knee to determine effectiveness of treatment.

[Technique #2 In LE Region Section \(Video 11.28\) in Nicholas & Nicholas](#)

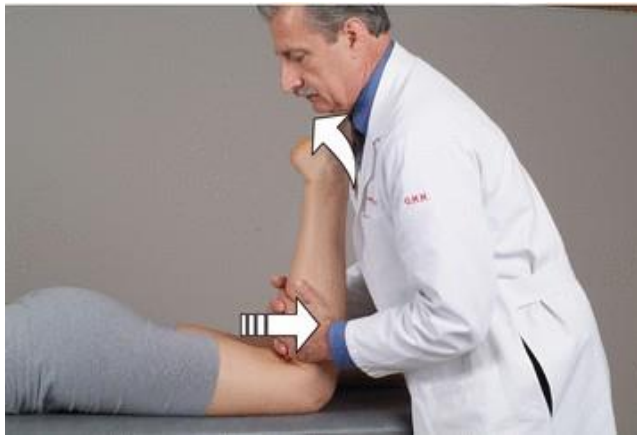
HVLA Posterior Tibia on Femur Somatic Dysfunction

1. The patient is asked to lie prone with the dysfunctional knee flexed to approximately 90 degrees if possible; the physician is standing or sitting at the foot of the table.
2. The dorsum of the patient's foot is placed on the anteromedial aspect of the physician's shoulder (this placement will induce plantar flexion of the foot, taking tension off the gastrocnemius muscle).
3. The physician interlaces and wraps their fingers around the proximal tibia just distal to the popliteal region as shown below:



g. 11.159

4. The physician delivers a thrust with both hands toward themselves and parallel to the table as follows:



g. 11.160

5. The physician reassesses posterior glide and the range of motion of the knee to determine effectiveness of treatment.

[Technique #3 in LE Region Section \(Video 11.29\) in Nicholas & Nicholas](#)

HVLA Anterior Dysfunction of the Fibular Head

Knee: Proximal Fibular Dysfunction

Short-Lever, Posterior with Inversion Emphasis

Ex: Right Anterior Fibular Head

Technique

1. The patient lies supine with a small pillow under the dysfunctional knee to maintain the knee in slight flexion.
2. The physician's caudad hand internally rotates the patient's ankle to bring the proximal fibula more anterior.
3. The physician places the heel of the cephalad hand over the anterior surface of the proximal



Steps 1 to 3 above

4. A thrust is delivered through the fibular head straight back toward the table (pulsed white arrow)



Step 4 above

5. Simultaneously, an internal rotation counterforce is introduced from the ankle (curved white arrow)
6. The physician reassesses the range of motion, position, and tenderness of the fibular head to determine the effectiveness of the technique

[Technique #5 in LE Region Section \(Video 11.30\) in Nicholas & Nicholas](#)

HVLA Posterior Dysfunction of the Fibular Head

Knee: Proximal Fibular Dysfunction

Long-Lever, Fulcrum/Eversion Emphasis

Ex: Left Posterior Fibular Head

Technique

1. The patient lies prone with the dysfunctional knee flexed at 90 degrees.
2. The physician stands at the side of the table opposite the side of the dysfunction
3. The physician places the MCP of the cephalad index finger behind the dysfunctional fibular head, and the hypothenar eminence is angled down into the hamstring musculature to form a wedge behind the knee
4. The physician's caudad hand grasps the ankle on the side of dysfunction and gently flexes the knee until the restrictive barrier is reached



Steps 1 to 4 above

5. The patient's foot and leg are gently externally rotated to carry the fibular head back against the fulcrum formed by the physician's cephalad hand (*white arrow below*)



6. The physician's caudad hand, controlling the patient's foot and ankle, delivers a thrust toward the patient's buttock in a manner that would normally result in further flexion of the knee (*white arrow below*). However, the wedge fulcrum formed by the physician's cephalad hand prevents any such motion.



Step 6

7. The physician reassesses the range of motion, position, and tenderness of the fibular head to determine the effectiveness of the technique.

[Technique #6 in LE Region Section \(Video 11.31\) in Nicholas & Nicholas](#)

Review Materials:

Sacral Diagnosis (Lab18) and Sacral MET (Lab 19 and Lab 20)

https://nyit.instructure.com/courses/24945/pages/ppom-i-required-videos-class-of-2027-2023-2024?module_item_id=749691

References:

- Nicholas A. & Nicholas E, *Atlas of Osteopathic Techniques*, Fourth edition, Lippincott Williams & Wilkins, 2021.
- OMM Year 2 Instructional Videos. NYITCOM OMM Department Media Library.
- Tank, P. W., Gest, T. R., Burkel, W. E., & Lippincott Williams & Wilkins, *Lippincott Williams & Wilkins Atlas of Anatomy*, Second edition, Wolters Kluwer Health/Lippincott Williams & Wilkins, 2019.